Machine-Learning-Assisted Analysis of a Static Relativistic Gravitational Potential: A Comparative Study with Newtonian Approximation and Error Quantification

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1 Abstract

Understanding the range of validity of classical gravitational models and identifying regimes where relativistic corrections become significant is an important problem in computational physics. In this work, we investigate a static relativistic gravitational potential through a combined analytical, numerical, and machine-learning-based approach.

The relativistic formulation is analyzed and compared with the corresponding Newtonian approximation through mathematical derivation and computational error analysis. The deviation between the two models is quantified using relative error measurements, allowing the transition between strong-field and weak-field gravitational regimes to be studied systematically.

To explore the capability of data-driven methods in approximating nonlinear physical systems, a computational dataset is generated from the analytical model and used to train multiple machine learning algorithms. Linear regression, polynomial regression, neural networks, and random forest regression are evaluated to compare their ability to reproduce the behavior of the relativistic potential.

The results demonstrate that machine learning models can effectively approximate the underlying physical relationship, with nonlinear approaches providing significantly improved performance over simpler regression methods. The predictions are interpreted as computational approximations that complement the analytical formulation rather than replacing the physical model.

This work presents a framework that combines theoretical physics, numerical analysis, and machine learning for studying nonlinear scientific systems. The approach highlights the potential of machine learning as a tool for computational modeling while maintaining consistency with established physical principles.

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2 Introduction

2.1 Motivation

Mathematical modeling is essential in studying physical systems because it enables the formulation of mathematical relationships that describe and predict physical phenomena. In fact, different theories in physics use mathematical relations that help represent and analyze physical behaviors.

Gravitational systems provide a prominent example where the limitations of different mathematical models become important. Although the classical theory of gravity proposed by Newton is accurate enough in most cases, it fails to properly describe physical processes when gravitational fields become much stronger [1]. Therefore, relativistic models based on Einstein's theory of gravity are required to study these systems and obtain accurate predictions [2].

The transition of a physical system from being described by one mathematical model to another is an intriguing computational task since it involves quantifying how one model deviates from another. This task will help to identify the limits of applicability of a classical model and find the areas where relativistic corrections should be used.

This study examines the problem using the methods of computation, including theoretical modeling, numerical analysis, and machine learning. The goal of the research is not to change the physical model itself, but to see how artificial intelligence can be applied in solving such problems.

2.2 Background of Relativistic Potential Models

Models for describing gravitational interaction have drastically changed from classical Newtonian mechanics to relativistic ones [1, 2]. The model of interaction from the Newtonian mechanics is described by forces between bodies. There is also gravitational potential that serves as an efficient tool for studying such a system.

Newtonian potential works effectively when gravitational field is weak, i.e., in weak-field approximation when it is safe to neglect the curving of spacetime around the object. Otherwise, there will be some deviations from this approach, and we need to consider a relativistic approach.

There are several relativistic models for describing the gravitational field [2]. These models differ from the Newtonian one because of some corrections arising from the interaction between matter, energy, and spacetime geometry. This difference becomes noticeable near the region of extremely strong gravitational fields. Therefore, the relativistic potential becomes different from the Newtonian one and the deviation is dependent on the scale of the system.

A numerical estimate of the difference between these models allows us to understand the limit beyond which the weak-field approximation becomes inaccurate. This allows us to investigate the range of applicability of this approximation.

In this paper, a relativistic static potential model will be investigated using mathematical analysis and computation [3]. The Newtonian approximation of the same model will be considered and the results of analysis will be presented through error estimations and approximation using machine learning techniques.

2.3 Computational Physics and Machine Learning

The use of computational techniques has become an integral component of modern scientific investigations, which makes it possible to study physical systems that may be difficult to analyze using purely analytical methods. Numerical simulation gives an opportunity to investigate mathematical models under different conditions and examine the dependencies between the parameters of physical systems.

But the traditional computational approaches are becoming more costly in case of multiple calculations in the case of the large size of parameter spaces. And therefore, the need for searching other types of approaches which would be able to learn and approximate complex mathematical relations has been emerged.

Machine learning offers a way of pattern recognition and building of predictive models based on data. Machine learning can serve as a surrogate modeling method in the scientific field, which allows building models that learn the dependencies between the input and output values (physical results of some theoretical system).

The application of machine learning in physics implies that apart from the need for accuracy there is a requirement of validation of the results in comparison with theoretical predictions and proper estimation of errors. Thus, the combination of machine learning with the analytical approaches and computational experiments is an important feature of the scientific machine learning investigations.

In this project, machine learning approaches will be applied to the static relativistic potential system in order to estimate whether it is possible to construct an accurate approximation to the physical relationship. The machine learning approach will be considered as a supplementary computational technique.

2.4 Objectives and Contributions

The primary goal of this paper is to develop a computational framework that combines theoretical physics and machine learning to model a static relativistic potential.

In particular, the study is aimed at exploring the connections between the relativistic and classical representations of gravitational phenomena and finding out if the machine learning models are capable of capturing the resulting physical connection.

The major goals of the current research include:

1. Developing the mathematical formulation of static relativistic potential and its classical approximation.

2. Evaluating the discrepancy between the relativistic and Newtonian models through error analysis.

3. Exploring the regime change (transition from the strong field to the weak field) through examining the behavior of the relative error.

4. Generating a computational data set based on the theoretical model for machine learning.

5. Testing several machine learning algorithms, such as regressions, neural networks, and ensembles.

6. Comparing the performance of different models through proper evaluations of their performance.

The main contribution of the current research is the combination of three components: analytical physics modeling, numerical error analysis, and machine learning approximation into one unified process. Rather than treating artificial intelligence as a replacement for traditional scientific approaches, this study explores its role as a complementary computational tool.

2.5 Organization of the Paper

The remainder of this paper is organized as follows. Section 2 presents the theoretical framework of the static relativistic potential and its Newtonian approximation. Section 3 describes the mathematical formulation and derivation of the error function. Section 4 presents the numerical analysis of the relativistic and classical models. Section 5 discusses computational implementation and dataset generation. Section 7 introduces the machine learning framework and describes the evaluated models. Section 8 presents the model performance results and analysis. Finally, Section 9 discusses the physical interpretation, limitations, and future extensions of the study.

3 Theoretical Framework

3.1 Physical Motivation

Mathematical models describing physical systems need to be accurate and reflect correctly the physical phenomena occurring within such systems depending on various conditions. While classical gravitational systems give reliable predictions in many cases, their limitations arise when dealing with strong enough gravitational fields and relativistic effects.

Comparison of different descriptions gives insight into the boundaries of applicability of one or another physical model. The comparison of the deviation between such models enables us to quantify the transition from one regime to another.

In the case of the weak-field approximation, the Newtonian approximation gives a good description, while relativistic corrections can be neglected. As the characteristic distance from the gravitational field decreases, relativistic corrections start to play a bigger role and the classical description becomes less precise.

The aim of this work is to examine the process of the transition from the classical to the relativistic description by constructing a computational framework which allows us to compare the relativistic and the Newtonian potentials.

By using analytical analysis, numerical calculations and machine learning algorithms we will investigate not only the physical characteristics of the system but also the possibility of obtaining computational approximations.

The theoretical framework proposed in this section serves as a basis for the subsequent mathematical description, analysis of errors and machine learning experiments.

3.2 Dimensionless Representation

In order to facilitate the mathematical manipulation of the model and generalize the outcome to various physical scales, the gravitational field is expressed in terms of a dimensionless variable.

The normalization of the radius vector is done using the characteristic length of the relativistic gravitational system:

$$x = \frac{r}{r_s}$$

where:

- r represents the radial distance from the gravitational source.
- r_s represents the characteristic relativistic scale of the system.
- x represents the normalized dimensionless radial parameter.

Dimensional variable gives several benefits. First, it eliminates dependence on the specific physical units and allows the system to be studied in terms of the relative distance instead of the absolute one.

The variable (x) can be also used for the examination of various gravitational regimes. The values of the variable (x) that are close to the characteristic length correspond to the relativistic gravitational systems. Larger values of the variable (x) correspond to gravitational systems with low intensity where classical approximation is a better choice.

In terms of this normalized distance both the relativistic and Newtonian potential models can be expressed, and thus the comparison between them will not depend on any specific physical system but will rely only on mathematics.

This dimensionless formulation forms the basis for the analytical derivations and computational simulations performed in the following sections.

3.3 Newtonian Approximation

The Newtonian gravitational potential provides the classical reference model for gravitational interactions in the weak-field limit. In this work, it is used as the baseline model against which the relativistic correction is quantified.

The Newtonian approximation assumes a fixed space-time framework and is accurate when gravitational fields are sufficiently weak. However, near strong-field regions, deviations from the Newtonian description become significant, motivating comparison with a relativistic formulation.

The mathematical definition and normalization of the Newtonian potential used in this work are presented in Section 4.2.

3.4 Static Relativistic Potential Model

To incorporate strong-field gravitational effects, this work considers a static relativistic potential derived from the Schwarzschild solution of general relativity.

Unlike the Newtonian approximation, the relativistic model introduces nonlinear corrections arising from spacetime curvature. These corrections become increasingly important near the Schwarzschild radius.

The complete mathematical formulation, normalization procedure, and dimensionless representation of the relativistic potential are described in Section 4.1.

3.5 Relativistic-Newtonian Comparison

The Newtonian and relativistic potential models describe the same gravitational system but differ in their treatment of gravitational effects.

The Newtonian potential provides a classical approximation that assumes weak gravitational fields and neglects relativistic corrections. In contrast, the relativistic potential incorporates nonlinear effects that arise from the relativistic description of gravity.

A comparison of the two models allows the accuracy of the classical approximation to be evaluated as a function of the normalized radial coordinate. In regions far from the characteristic relativistic scale, the two potentials exhibit similar functional behavior, allowing Newtonian gravity to provide a qualitative approximation. However, quantitative comparison through relative error analysis is required to determine the accuracy of this approximation.

Closer to the strong-field region, however, the difference between the models becomes increasingly significant. This deviation reflects the growing importance of relativistic corrections and motivates the use of quantitative error analysis.

To measure the discrepancy between the two descriptions, a relative error function is introduced in the following section. This function provides a numerical basis for evaluating the validity range of the Newtonian approximation and identifying the transition between classical and relativistic gravitational behavior.

4 Mathematical Formulation

4.1 Relativistic Potential Equation

The relativistic gravitational potential investigated in this study is derived from the Schwarzschild solution and is expressed as

$$\Phi_{GR}(r) = \frac{c^2}{2}\left(\sqrt{1 - \frac{2GM}{rc^2}} - 1\right).$$

where:

- G represents the gravitational constant.
- M represents the mass of the gravitating source.
- r represents the radial distance.
- c represents the speed of light.

Introducing the Schwarzschild radius

$$r_s = \frac{2GM}{c^2}$$

and the dimensionless coordinate

$$x = \frac{r}{r_s},$$

Substituting:

$$GM = \frac{r_s c^2}{2}$$

and

$$r = x r_s,$$

we obtain:

$$\Phi_{GR}(x) = -\frac{c^2}{2}\left(\sqrt{1 - \frac{1}{x}} - 1\right).$$

the dimensionless Relativistic potential is obtained by normalizing with c^2:

$$\phi_{GR}(x) = \frac{\Phi_{GR}}{c^2} = -\frac{1}{2}\left(\sqrt{1 - \frac{1}{x}} - 1\right)$$

This equation represents the reference relativistic model used throughout the study. The nonlinear square-root dependence introduces deviations from classical gravity that become significant near the Schwarzschild radius.

For large values of (x), the relativistic potential follows the same inverse-distance dependence as the Newtonian approximation. However, because the leading-order coefficients differ, the two potentials do not converge to the same normalized value. The difference arises because the adopted relativistic potential is normalized through the Schwarzschild metric expression rather than constructed as the classical Newtonian limit; therefore the comparison is performed within the defined potential formulation. This difference is quantified through the relative error analysis performed in later sections. Therefore, the weak-field limit in this formulation should be interpreted as convergence in functional behavior rather than exact numerical agreement between the two normalized potentials.

4.2 Newtonian Potential Equation

Newtonian Potential Equation The Newtonian gravitational potential is taken as the classical reference potential for comparison purposes with the relativistic potential equation. The potential equation represents the gravitational effect of the mass source assuming that the effects of spacetime and the gravitational field are in the weak-field approximation region.

The standard Newtonian gravitational potential is given by:

$$\Phi_N(r) = -\frac{GM}{r}$$

where:

- G represents the gravitational constant.
- M represents the mass of the gravitational source.
- r represents the radial distance from the source.

Using the dimensionless radial coordinate introduced previously:

$$x = \frac{r}{r_s}$$

and the Schwarzschild radius

$$r_s = \frac{2GM}{c^2}$$

the Newtonian potential can be rewritten as:

$$\Phi_N(x) = -\frac{c^2}{2x}$$

the normalized Newtonian potential becomes:

$$\phi_N(x) = \frac{\Phi_N}{c^2} = -\frac{GM}{rc^2}$$

Substituting:

$$GM = \frac{r_s c^2}{2}$$

and

$$r = x r_s,$$

we obtain:

$$\phi_N(x) = -\frac{1}{2x}$$

This normalized expression provides a simplified representation that allows direct comparison with the relativistic potential model.

The Newtonian approximation serves as the standard for which the relativistic corrections are taken into consideration. For large values of (x), both models exhibit similar inverse-distance behavior. However, the relative error analysis is required to determine the actual quantitative agreement between the two descriptions. However, as (x) goes deeper into stronger-field regions, the relativistic potential starts to differ from the classical approximation.

4.3 Derivation of Relative Error Expression

To quantify the difference between the relativistic and Newtonian descriptions, the relative error between the two potential models is calculated.

The relative error provides a measure of how much the Newtonian approximation deviates from the relativistic potential at a given value of the normalized radial coordinate (x).

The general form of the relative error is defined as:

$$\epsilon(x) = \left| \frac{\phi_{GR}(x) - \phi_N(x)}{\phi_{GR}(x)} \right| \times 100$$

where:

- $\phi_{GR}(x)$ represents the normalized relativistic potential,
- $\phi_N(x)$ represents the normalized Newtonian approximation,
- $\epsilon(x)$ represents the percentage deviation between the two models.

Substituting the expressions for the relativistic and Newtonian potentials into the error equation gives a mathematical relationship describing the accuracy of the classical approximation across different gravitational regimes.

This error function allows the transition between relativistic and classical behavior to be studied quantitatively. When $\epsilon(x)$ is small, the Newtonian model provides a close approximation to the relativistic system. However, an increasing value of $\epsilon(x)$ indicates that relativistic corrections become significant.

The resulting error function is evaluated computationally over a range of (x) values to analyze the behavior of the approximation and identify regions where classical gravity remains sufficiently accurate.

This formulation provides the connection between the theoretical model and the numerical analysis performed in this study.

4.4 Simplified Error Formula

After substituting the relativistic and Newtonian potential expressions into the relative error equation, the resulting expression can be simplified to obtain a direct relationship between the error and the normalized radial coordinate (x).

The simplified form obtained is:

$$\epsilon(x) = 100 \left| \frac{-x^{3/2} + \sqrt{x} + x\sqrt{x-1}}{x(-\sqrt{x} + \sqrt{x-1})} \right|$$

This expression represents the percentage deviation between the relativistic potential and the Newtonian approximation as a function of the dimensionless radial coordinate.

The equation provides a direct way to analyze how the accuracy of the Newtonian model changes across different gravitational regimes. As (x) increases, the relativistic corrections become smaller and the error approaches the weak-field limit, where the Newtonian approximation becomes increasingly accurate.

Conversely, for values of (x) closer to the strong-field region, the deviation between the two models becomes more significant. This behavior reflects the increasing influence of relativistic effects as the system moves away from the conditions where classical gravity provides a reliable approximation.

The simplified error expression is particularly useful because it removes the need to repeatedly compare the two potential functions separately. Instead, the accuracy of the Newtonian approximation can be directly evaluated from a single mathematical relationship.

This analytical formulation is later used for numerical error analysis, including the determination of accuracy thresholds and the identification of the transition region between relativistic and classical gravitational behavior.

5 Error Analysis

5.1 Error Behaviour

The relative error function derived in the previous section provides a quantitative measure of the difference between the relativistic potential and the Newtonian approximation. By evaluating this function over a range of values of the normalized radial coordinate (x), the accuracy of the classical model can be studied.

The behavior of the error is expected to depend strongly on the gravitational regime being considered. In regions closer to the characteristic relativistic scale, the effects of relativistic corrections become more significant, resulting in a larger deviation between the two potential models.

As the value of (x) increases, the gravitational field becomes weaker and the functional behavior of the relativistic and Newtonian descriptions becomes increasingly similar. However, due to the difference in their asymptotic normalization, the relative error does not decrease to zero and instead approaches a limiting value.

The variation of the relative error with respect to the normalized radial coordinate is shown in Figure 1.

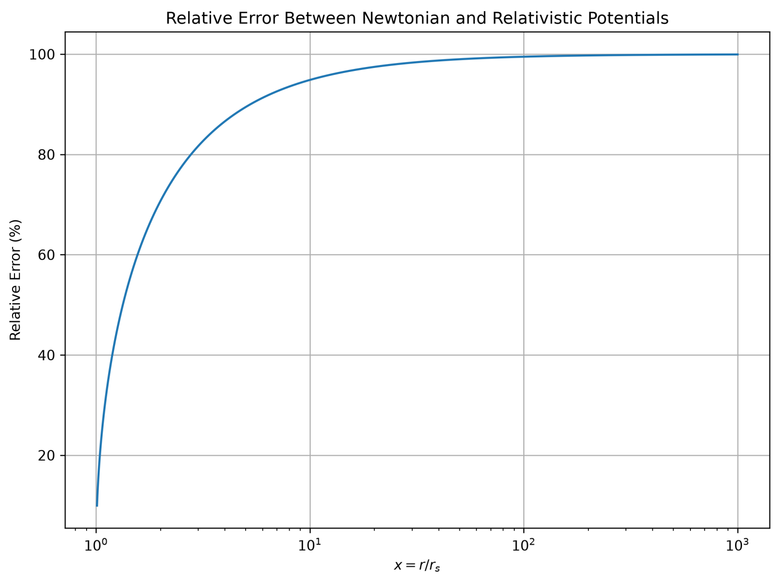

Figure 1: Relative error between the normalized relativistic potential and the Newtonian approximation as a function of the dimensionless radial coordinate x. The minimum error occurs near $x = 1.01$, while the error increases toward the weak-field regime due to the differing asymptotic normalizations of the two models.

This behavior demonstrates the transition from a region where relativistic corrections significantly modify the potential toward a region where both models exhibit similar qualitative inverse-distance behavior. However, due to the difference in asymptotic normalization, the Newtonian model does not achieve vanishing relative error under the chosen formulation.

The numerical evaluation of the error function allows this transition to be examined quantitatively rather than only through theoretical assumptions. By measuring how the error changes with (x), the range in which Newtonian gravity achieves a specified accuracy can be determined.

The error analysis therefore provides an important link between the theoretical formulation and the quantitative evaluation of the classical approximation within the considered model.

5.2 Error Threshold Analysis

A numerical threshold analysis was performed to determine whether the Newtonian approximation reaches commonly used accuracy levels within the sampled domain.

The threshold condition was defined as:

$$\epsilon(x) \leq \epsilon_{limit}$$

where ϵ_{limit} represents a specified maximum relative error.

Three thresholds were investigated:

- 10% relative error,
- 5% relative error,
- 1% relative error.

The numerical evaluation showed that the relative error reaches its minimum value near the lower boundary of the investigated domain:

$$x = 1.01$$

with a corresponding relative error of approximately:

$$\epsilon = 9.950372$$

However, the analysis further revealed that the relative error never falls below either the 5% or 1% thresholds within the investigated domain:

$$1.01 \leq x \leq 1000$$

The minimum observed relative error across the sampled dataset was approximately 9.95%.

These results indicate that, for the specific relativistic potential considered in this study, the Newtonian approximation does not achieve better than approximately 10% accuracy within the investigated domain. The minimum relative error occurs near the lower boundary of the sampled range (x = 1.01), while the discrepancy increases as x becomes larger.

This behavior arises from the different asymptotic forms of the two potentials. In the large-distance limit, the relativistic potential approaches $\phi_{GR} \approx -1/(4x)$, whereas the Newtonian approximation follows $\phi_N = -1/(2x)$. Consequently, the relative error increases toward 100% as $x \to \infty$ rather than remaining near 10%. Therefore, the observed 9.95% value represents the minimum error within the chosen domain, not the asymptotic error behavior.

5.3 Strong-Field and Weak-Field Transition

The variation of the relative error with the normalized radial coordinate provides a way to identify the transition between strong-field and weak-field gravitational regimes.

In the strong-field region, where (x) approaches the characteristic relativistic scale, the assumptions behind the Newtonian approximation become less accurate. The contribution of relativistic corrections becomes more significant, leading to a larger difference between the relativistic and classical potential models.

As the value of (x) increases, the influence of relativistic corrections gradually decreases. The system moves toward the weak-field regime, where the Newtonian approximation becomes increasingly consistent with the relativistic description.

This transition can be understood through the behavior of the relative error. A larger error indicates that the classical approximation does not fully capture the physical behavior of the system, while a smaller error indicates that the Newtonian model provides an effective approximation.

By analyzing the points where the error crosses predefined accuracy limits, the boundary between these regimes can be estimated quantitatively. Instead of defining the transition only through theoretical assumptions, this approach provides a measurable criterion based on the accuracy of the approximation.

The identification of this transition region is important because it establishes the range in which classical gravitational models remain applicable and the range where relativistic corrections become necessary. This analysis also provides a physically meaningful interpretation of the dataset later used for machine learning experiments.

5.4 Asymptotic Behaviour

The asymptotic behavior of the relativistic and Newtonian potentials provides important insight into their relationship in the weak-field regime.

As the normalized radial coordinate (x) increases, both potentials approach an inverse-distance dependence, indicating similar qualitative behavior. However, they do not converge to the same value because their leading-order terms differ:

$$\phi_{GR} \approx -\frac{1}{4x}$$

whereas

$$\phi_N = -\frac{1}{2x}.$$

Therefore, the Newtonian approximation captures the general distance-dependent trend of the relativistic potential but retains a systematic quantitative discrepancy due to the difference in normalization. The observed asymptotic discrepancy is a consequence of the chosen normalization of the relativistic potential and should not be interpreted as a failure of the Newtonian weak-field approximation itself.

Numerical evaluation of the relative error reveals that the minimum observed deviation within the investigated domain occurs near the lower boundary:

$$\epsilon_{min} = 9.950372\%$$

at approximately:

$$x = 1.01.$$

The difference between the relativistic and Newtonian potentials is shown in Figure 2, illustrating the systematic deviation between the two models.

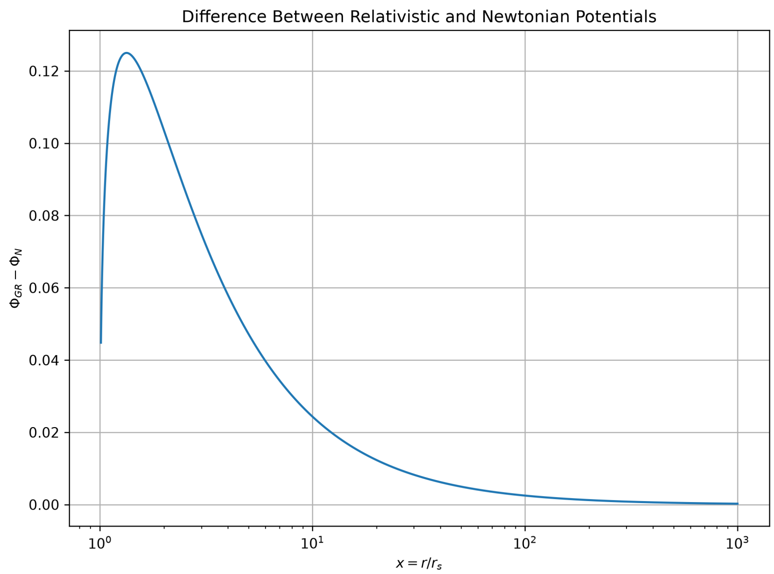

Figure 2: Difference $\Delta\phi = \phi_{GR} - \phi_N$ between the normalized relativistic and Newtonian potential models across the investigated domain. The plot illustrates the systematic deviation between the two formulations.

However, this minimum value should not be interpreted as the asymptotic error. As x increases, the relative error grows and approaches 100% in the large-distance limit because the ratio between the leading-order terms remains different.

This behavior explains why the 5% and 1% accuracy thresholds are never reached within the sampled range:

$$1.01 \leq x \leq 1000.$$

The location of the maximum deviation is further illustrated in Figure 3.

Therefore, while the Newtonian approximation reproduces the qualitative inverse-distance behavior of the relativistic potential, explicit error analysis reveals a persistent and increasing quantitative discrepancy. This result demonstrates the importance of evaluating approximation accuracy through numerical comparison rather than assuming convergence solely from similar functional behavior.

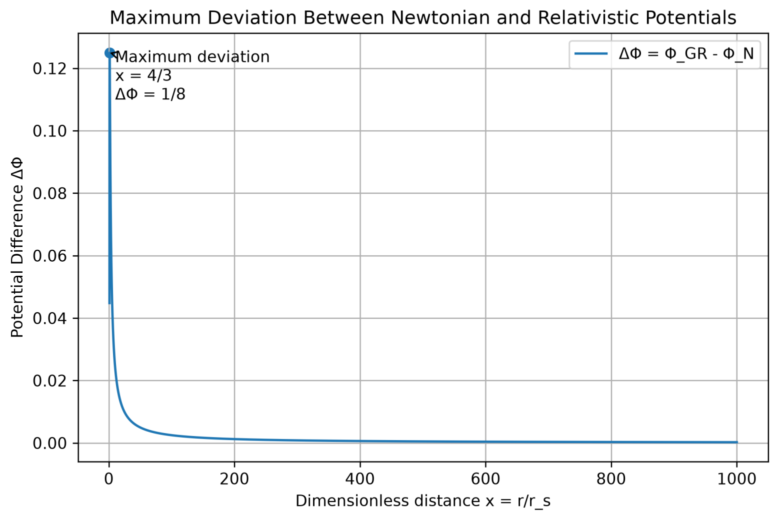

Figure 3: Annotated plot of the potential difference showing the location of maximum deviation between the relativistic and Newtonian models.

6 Gravitational Field Analysis

6.1 Newtonian Field

The gravitational field is obtained from the spatial variation of the potential. In the Newtonian framework, the gravitational field is defined as the negative gradient of the gravitational potential:

$$g_N = -\frac{d\Phi_N}{dr}$$

Using the Newtonian potential:

$$\Phi_N(r) = -\frac{GM}{r}$$

the corresponding gravitational field becomes:

$$g_N = \frac{GM}{r^2}$$

This represents the classical inverse-square behavior of gravity, where the field strength decreases with increasing distance from the source.

Using the normalized coordinate (x), the Newtonian field can be analyzed in terms of the relative distance scale rather than absolute units. This allows direct comparison with the relativistic field representation.

The Newtonian field provides the classical reference behavior for evaluating the influence of relativistic corrections. In the weak-field regime, it is expected to closely match the relativistic prediction, while deviations become more noticeable as the system approaches stronger gravitational regions.

The analysis of the Newtonian field therefore provides the baseline required for understanding how relativistic effects modify the gravitational behavior of the system.

6.2 Relativistic Field

In the relativistic description, the gravitational field is obtained from the variation of the relativistic potential with respect to the radial coordinate:

$$g_{GR} = -\frac{d\Phi_{GR}}{dr}$$

Unlike the Newtonian formulation, the relativistic field includes corrections that arise from the modified structure of the gravitational potential. These corrections become increasingly important in regions where the gravitational field is stronger.

The behavior of the relativistic field differs from the Newtonian prediction mainly due to the nonlinear dependence introduced by the relativistic formulation. At larger values of (x), where the gravitational field is weaker, the relativistic field gradually approaches the Newtonian result. However, closer to the strong-field region, the difference between the two descriptions becomes more significant.

The comparison of these two field models provides additional information beyond the potential analysis. While the potential comparison measures the difference in gravitational energy representation, the field comparison examines how the gravitational influence itself changes across different regimes.

This analysis helps establish a more complete understanding of the system by considering both the potential and the resulting field behavior before applying computational and machine learning methods.

6.3 Comparative Analysis

The comparison between the Newtonian and relativistic gravitational fields provides a broader perspective on the transition between classical and relativistic descriptions.

In the weak-field regime, the difference between the two models becomes minimal, indicating that Newtonian gravity provides an accurate approximation of the relativistic behavior. This agreement explains why classical gravitational models remain effective for many practical applications where relativistic corrections are extremely small.

However, as the system moves toward stronger gravitational regions, the relativistic corrections become increasingly important. The deviation between the two field descriptions reflects the limitations of the classical approximation and highlights the need for a relativistic treatment.

The combined analysis of potential and field behavior demonstrates that the transition from Newtonian to relativistic behavior is gradual rather than abrupt.

The accuracy of the classical approximation depends on the physical regime being considered and can be quantified through computational error analysis.

This comparison provides additional validation for the dataset generation process used in this study. Since the machine learning models are trained on values derived from the relativistic formulation, understanding the physical behavior of the underlying system is essential for interpreting the predictions produced by the models.

7 Dataset Construction

7.1 Data Generation

The dataset used in this study was generated directly from the mathematical formulation of the static relativistic potential model. Instead of relying on externally collected data, the values were produced by evaluating the theoretical equations across a defined range of the normalized radial coordinate (x).

For each value of (x), the corresponding relativistic potential value was calculated and stored as the target physical quantity for machine learning analysis. The Newtonian approximation and error-related quantities were also evaluated to support comparison and validation.

The generated dataset represents the relationship between the input parameter and the resulting gravitational potential behavior. This approach ensures that the machine learning models are trained on data that follows the underlying physical laws represented by the mathematical model.

The dataset generation process can be summarized as:

1. Selecting a range of normalized radial coordinate values (x).
2. Evaluating the relativistic potential model for each value.
3. Storing the resulting input-output relationship.
4. Using the generated dataset for training and evaluating machine learning models.

By generating the dataset from the analytical model, the study creates a controlled environment where machine learning performance can be evaluated against a known physical relationship.

7.2 Dataset Range and Sampling

To analyze the behavior of the relativistic potential across different gravitational regimes, the dataset was generated over multiple ranges of the normalized radial coordinate (x).

Two different sampling ranges were considered in this study:

1. A focused dataset covering a smaller range of (x), designed to capture the behavior closer to the stronger-field region where relativistic effects are more significant.

2. A broader dataset extending to larger values of (x), allowing the analysis of the weak-field regime where the relativistic potential approaches the Newtonian approximation.

The smaller-range dataset was used for detailed analysis of the region where the potential changes more rapidly, while the extended-range dataset provided a wider representation of the overall behavior of the system.

Sampling across these ranges allows the machine learning models to learn both the nonlinear behavior present in stronger gravitational regions and the gradual convergence toward classical behavior at larger distances.

The choice of sampling range is important because machine learning models learn patterns from the provided data distribution. A dataset that includes both strong-field and weak-field behavior allows the trained models to develop a more complete representation of the underlying physical relationship.

This approach ensures that the computational experiments evaluate not only the predictive capability of machine learning models but also their ability to approximate physical behavior across different regimes.

7.3 Statistical Properties

Before applying machine learning algorithms, the generated dataset was analyzed to understand its structure and characteristics. Since the dataset is derived from a mathematical model, its statistical properties directly reflect the behavior of the underlying physical system.

The primary input variable is the normalized radial coordinate:

$$x = \frac{r}{r_s}$$

which represents the position parameter of the system. The corresponding target variable is the relativistic potential:

$$\phi_{GR}(x)$$

which the machine learning models are trained to approximate.

The relationship between these variables is nonlinear, particularly in regions where relativistic effects become significant. This nonlinear behavior provides a suitable test case for evaluating different machine learning approaches, ranging from simple regression models to more flexible nonlinear models.

The dataset characteristics are important because the performance of a machine learning model depends strongly on the complexity and distribution of the available data. Regions with rapid changes in the potential require models capable of capturing nonlinear patterns, while smoother regions provide a simpler approximation problem.

The generated dataset therefore serves as a controlled scientific benchmark where the accuracy of machine learning models can be evaluated against a known physical relationship.

The combination of theoretical generation and statistical analysis ensures that the machine learning results can be interpreted in the context of the original gravitational model rather than as purely data-driven predictions.

The generated dataset contains 10,000 samples spanning the normalized radial coordinate range from x = 1.01 to x = 1000. Descriptive statistical analysis was performed to characterize the dataset prior to machine learning training.

The mean value of the normalized radial coordinate was 144.86, while the mean relativistic potential was −0.04379. The relative error between the Newtonian and relativistic models ranged from approximately 9.95% to 99.95%, with an average value of 91.24%.

These statistics indicate that significant deviations exist between the classical and relativistic descriptions throughout much of the sampled range. The wide variation of the potential values and error measurements provides a suitable benchmark for evaluating the approximation capabilities of different machine learning models.

8 Machine Learning Framework

8.1 Dataset Generation and Preparation

The numerical analysis and machine learning experiments were performed using separate datasets designed for their respective purposes.

For analytical error evaluation, threshold analysis, and asymptotic behaviour, a dataset spanning:

$1.01 \leq x \leq 1000$

was generated. This range allows investigation of both strong-field and weak-field regimes.

For machine learning experiments, a restricted dataset:

$1.01 \leq x \leq 100$

was used. At very large values of x, the relativistic potential approaches a slowly varying weak-field limit, causing reduced variation in the target function and making the learning problem less informative. Therefore, the reduced range was selected to provide a more meaningful representation of the nonlinear structure of the potential.

The machine learning models were trained and evaluated on this dataset using an 80:20 train-test split. The dataset consists of the normalized radial coordinate x as the input feature and the relativistic potential

$$\phi_{GR}(x)$$

as the target variable. Additional quantities such as the Newtonian potential, potential difference, and relative error were retained for numerical analysis but were not used as machine learning inputs.

8.2 Problem Definition

The objective of the machine learning component in this study is to develop computational models capable of approximating the behavior of the static relativistic potential using data generated from the theoretical formulation.

The problem can be considered a supervised learning task, where the input variable is the normalized radial coordinate (x), and the target output is the corresponding relativistic potential value:

$$x \to \phi_{GR}(x)$$

The purpose of the machine learning models is to learn an efficient approximation of the relationship between the input parameter and the physical quantity, allowing rapid prediction without repeatedly evaluating the original analytical expression.

Several machine learning approaches with different levels of complexity are evaluated to understand how model architecture influences approximation accuracy. The models include traditional regression methods, nonlinear learning approaches, and ensemble-based techniques.

The selected models represent different learning strategies:

- Linear regression, which provides a simple baseline model.
- Polynomial regression, which introduces nonlinear feature relationships.
- Neural networks, which can learn complex nonlinear patterns.
- Random forest regression, which uses an ensemble of decision-based models to capture relationships within the data.

Each model is evaluated using standard performance metrics and compared against the known physical solution to determine its ability to approximate the relativistic potential.

This framework allows the study to investigate not only whether machine learning can reproduce the physical relationship, but also how different algorithms perform when applied to a scientifically generated dataset.

All machine learning models were implemented using the Scikit-learn library [4].

8.3 Linear Regression

Linear regression is used as the first baseline model to evaluate how well a simple mathematical approximation can represent the behavior of the relativistic potential.

The model assumes a linear relationship between the input variable and the target output, represented as:

$$\hat{y} = ax + b$$

where:

- $\hat{y}$ represents the predicted potential value.
- x represents the normalized radial coordinate.
- a and b are parameters learned during training.

Although the actual relationship between (x) and $\phi_{GR}(x)$ is nonlinear, linear regression provides a useful reference point for understanding the complexity of the learning problem.

The purpose of including this model is not to achieve the highest prediction accuracy, but to establish a performance baseline. If a simple linear relationship performs poorly, it indicates that the underlying physical system contains nonlinear behavior that cannot be captured using a basic approximation.

The trained linear regression model is evaluated using standard regression metrics and compared with more advanced models in later sections.

This comparison helps demonstrate how increasing model complexity affects the ability of machine learning methods to approximate the relativistic potential.

Regression methods are widely used as baseline approaches for approximating relationships between variables in statistical learning [5, 6].

8.4 Polynomial Regression

Polynomial regression is introduced to investigate whether adding nonlinear terms improves the approximation of the relativistic potential compared to a simple linear model.

Unlike linear regression, which assumes a direct proportional relationship between the input and output, polynomial regression expands the input variable into higher-order terms:

$$\hat{y} = a_0 + a_1 x + a_2 x^2 + ... + a_n x^n$$

where the coefficients $a_0, a_1, \ldots, a_n$ are learned from the training data.

By including additional powers of (x), the model gains the ability to represent more complex curves and approximate nonlinear relationships. This makes polynomial regression a natural intermediate approach between simple linear models and more flexible machine learning architectures.

However, increasing polynomial complexity can introduce challenges such as overfitting, where the model learns the specific characteristics of the training data but performs poorly on unseen examples.

In this study, polynomial regression is used to evaluate whether a mathematically enhanced regression approach can capture the nonlinear behavior of the relativistic potential. Its performance is compared with both simpler models and more advanced learning techniques to understand the advantages and limitations of different approaches.

The results from this model provide insight into whether the complexity of the physical relationship can be effectively represented through traditional mathematical transformations or requires more flexible machine learning architectures.

8.5 Neural Network

A neural network model is introduced to evaluate whether a more flexible learning architecture can capture the nonlinear relationship present in the relativistic potential system.

Unlike traditional regression methods, neural networks can learn complex relationships by combining multiple layers of interconnected computational units. The theoretical foundation of neural networks and deep learning methods is described in modern machine learning literature [7]. The general structure of a neural network consists of an input layer, one or more hidden layers, and an output layer.

In this study, the input to the network is the normalized radial coordinate:

$$x$$

and the output is the predicted relativistic potential:

$$\hat{\phi}_{GR}(x)$$

The model learns an approximation function:

$$f(x) \approx \phi_{GR}(x)$$

during the training process by adjusting internal parameters to minimize the difference between predicted and actual values.

The advantage of neural networks in this problem is their ability to represent nonlinear relationships without requiring an explicitly defined mathematical form. Since the relativistic potential contains nonlinear behavior, this flexibility allows the model to capture patterns that simpler regression methods may fail to represent.

The performance of the neural network is evaluated using regression metrics and compared with other models. The objective is not to replace the analytical formulation, but to investigate whether a learned approximation can reproduce the behavior of the physical model with high accuracy.

The results demonstrate the capability of neural networks as a computational approximation method for nonlinear scientific systems and provide a comparison between data-driven modeling and analytical approaches.

8.6 Random Forest

Random forest regression is included in this study as an ensemble-based machine learning approach for approximating the relativistic potential.

A random forest consists of multiple decision trees that are trained on different subsets of the data. The random forest approach follows the ensemble learning methodology introduced by Breiman [8]. Each individual tree learns a set of decision-based relationships between the input variable and the target output, and the final prediction is obtained by combining the outputs of all trees.

The prediction process can be represented as:

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} T_i(x)$$

where:

- $T_i(x)$ represents the prediction from the i^{th} decision tree.
- N represents the total number of trees.
- $\hat{y}$ represents the final predicted output.

The advantage of random forest regression is its ability to model nonlinear relationships while reducing the risk of overfitting compared to a single decision tree. Since the relativistic potential exhibits nonlinear behavior, an ensemble of decision-based models can effectively capture variations within the generated dataset.

In this study, the random forest model is evaluated alongside regression-based methods and neural networks to compare different learning strategies. Its performance provides insight into how ensemble learning approaches handle physically generated nonlinear datasets.

The high predictive accuracy achieved by the model demonstrates that machine learning techniques can approximate the numerical behavior of the relativistic potential when sufficient and representative training data are provided. However, the predictions remain dependent on the quality and range of the underlying physical dataset. Since the training data is generated from the analytical model, the machine learning performance represents the ability of the algorithms to reproduce the known physical relationship rather than independently derive the underlying law.

8.7 Feature Importance Analysis

In addition to evaluating prediction accuracy, understanding the influence of input variables is an important aspect of interpreting machine learning models.

Feature importance analysis was included as an interpretability check to confirm that the machine learning model relies on the physically meaningful input parameter used in the formulation. In this study, the primary feature considered is the normalized radial coordinate:

$$x = \frac{r}{r_s}$$

Since the dataset is generated from a single physical variable, the analysis focuses on understanding how changes in (x) influence the predicted relativistic potential.

For tree-based models such as random forest regression, feature importance can be calculated by analyzing how much each feature contributes to reducing prediction uncertainty during the learning process.

The feature importance analysis provides an additional interpretation layer by connecting the machine learning model back to the underlying physics. Rather than treating the model as a complete black box, this approach helps verify that the learned relationship is consistent with the expected dependence of the potential on the radial coordinate.

Although the present system contains a single dominant input variable, this analysis establishes a foundation for future extensions involving multiple physical parameters, where interpretability becomes increasingly important.

The combination of prediction performance and model interpretation allows the machine learning results to be evaluated from both computational and scientific perspectives.

9 Results and Analysis

9.1 Model Performance Comparison

The performance of the machine learning models was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

The obtained results are summarized below.

Table 1: Performance comparison of machine learning models.

Model	MAE	RMSE	R^2
Linear Regression	0.05579	0.07709	0.26914
Polynomial Regression	0.05624	0.07908	0.23091
Neural Network	0.00745	0.01038	0.98347
Random Forest	0.000028	0.000084	0.999999

The comparison between the analytical relativistic potential and machine learning predictions is visualized in Figure 4.

The results show a clear difference in predictive performance among the evaluated models. Linear and polynomial regression achieved relatively poor performance, indicating that the relationship between the normalized radial coordinate and the relativistic potential is strongly nonlinear.

The neural network significantly improved prediction accuracy, achieving an R^2 value above 0.98. This demonstrates its ability to capture the nonlinear structure of the generated physical dataset.

The random forest model produced the best overall performance, achieving an almost perfect fit with an R^2 value of approximately 0.999999 while maintaining extremely small prediction errors. The ensemble-based structure of the model allowed it to accurately approximate the relativistic potential within the machine

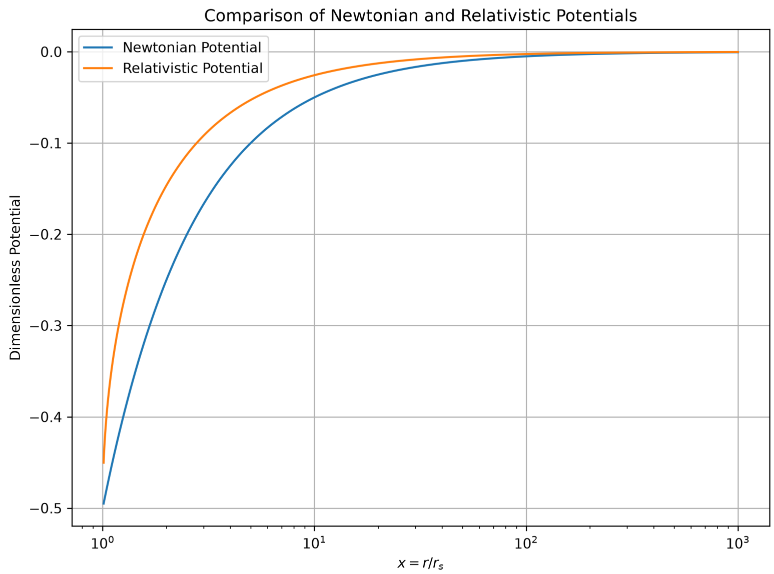

Figure 4: Comparison of machine learning predictions with the analytical relativistic potential. The figure illustrates the ability of different models to reproduce the theoretical relationship.

learning dataset range used in this study. The extremely high performance of the random forest model is expected because the dataset is generated directly from a deterministic analytical function with a single input variable.

These results indicate that nonlinear machine learning approaches are substantially more effective than traditional regression methods for modeling the relativistic potential considered in this study.

9.1.1 Numerical Error Analysis Summary

The computational analysis produced the following key numerical results:

$$\Delta\phi = \phi_{GR} - \phi_N$$

The threshold analysis showed that the minimum relative error occurs at the lower boundary of the investigated domain (x = 1.01,), where the error is approximately 9.95%. The 5% and 1% accuracy thresholds were not reached anywhere within the sampled domain. As x increases, the relative error grows and approaches approximately 100% due to the different asymptotic behavior of the relativistic and Newtonian potentials.

Table 2: Summary of numerical error analysis results.

Quantity	Value
Number of samples	10,000
Numerical Analysis Domain	$1.01 \leq x \leq 100$
Maximum deviation location	$x = \frac{4}{3}$
Maximum deviation	$\Delta\phi_{\mathrm{max}} = \frac{1}{8}$
Minimum relative error	9.950372%
Maximum relative error	99.949987% (approaching 100% as $x \to \infty$)

9.2 Prediction Analysis

The prediction results were further analyzed by comparing the machine learning outputs with the original relativistic potential values generated from the analytical model.

The prediction plots provide a visual representation of how closely each model follows the true physical relationship between the normalized radial coordinate (x) and the relativistic potential $\phi_{GR}(x)$.

The results show that simpler regression models exhibit noticeable deviations from the actual curve, particularly in regions where the relationship becomes more nonlinear. This behavior is expected because these models rely on limited assumptions about the structure of the underlying function.

The neural network model demonstrates improved agreement with the analytical solution by successfully capturing the nonlinear variation of the potential across the input range. The smoother representation obtained from the neural network reflects its ability to learn complex patterns from the training data.

The random forest model produces predictions that closely match the original relativistic potential values throughout the evaluated machine learning domain. This indicates that the ensemble-based approach is highly effective at approximating the numerical relationship present in the generated dataset.

The prediction capability of the random forest model is shown in Figure 5.

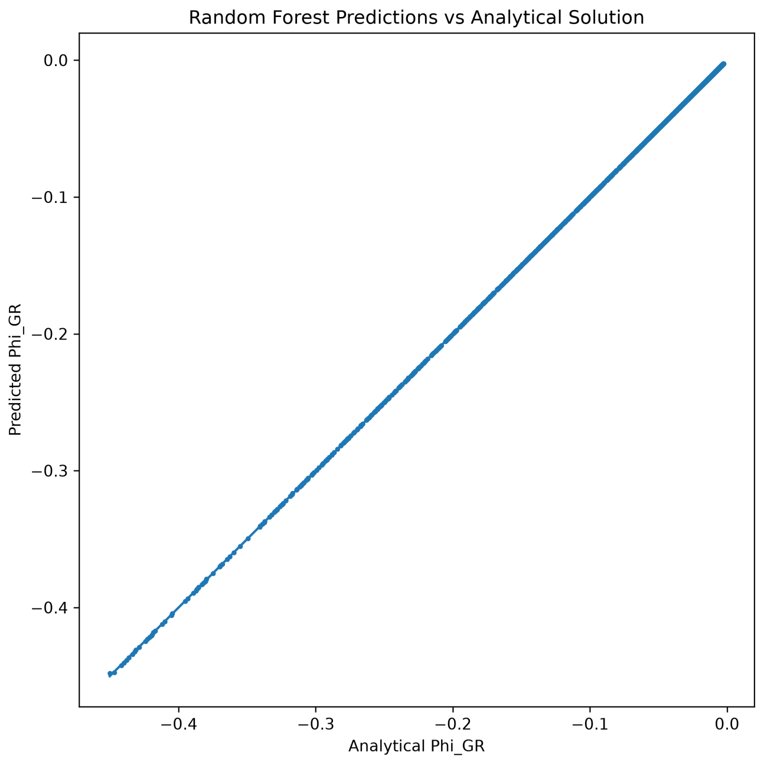

Figure 5: Random forest regression predictions compared with the analytical relativistic potential over the machine-learning dataset range.

The prediction analysis confirms the trends observed from the performance metrics. Models with greater capacity to capture nonlinear behavior provide more accurate approximations of the relativistic potential.

However, these results should be understood as an evaluation of machine learning as an approximation technique for a known physical model. The analytical formulation remains the source of the physical relationship, while the machine learning models provide computational representations of that relationship.

9.3 Residual Analysis

To further evaluate the performance of the machine learning models, the prediction errors were analyzed through residual analysis.

The residual is defined as the difference between the true relativistic potential value and the value predicted by the machine learning model:

$$Residual = \phi_{GR}(x) - \hat{\phi}_{GR}(x)$$

where:

- $\phi_{GR}(x)$ represents the analytical relativistic potential,
- $\hat{\phi}_{GR}(x)$ represents the machine learning prediction.

Analyzing residuals provides additional information about model behavior that cannot be obtained from overall accuracy metrics alone. A model with randomly distributed small residuals generally indicates that it has successfully captured the underlying relationship, while systematic patterns in residuals may indicate limitations in the model.

For simpler regression approaches, larger residual variations are expected due to their limited ability to represent the nonlinear structure of the relativistic potential. These deviations highlight the difficulty of approximating the physical relationship using restricted mathematical assumptions.

The neural network and random forest models demonstrate significantly smaller residuals compared with the simpler regression approaches, indicating improved approximation capability. The reduction in prediction error suggests that these models are better able to capture the nonlinear dependence between the normalized radial coordinate and the relativistic potential.

Residual analysis also provides insight into the reliability of machine learning predictions across different regions of the dataset. Regions with larger deviations can indicate areas where the underlying physical relationship is more difficult for the model to approximate.

This analysis complements the performance metrics and prediction plots by providing a more detailed understanding of how each model represents the relativistic potential.

The residual distribution provides additional information about prediction accuracy and model behavior.

9.4 Feature Importance Results

Feature importance analysis was performed to examine how the input variable contributes to the machine learning predictions.

In the present study, the dataset contains a primary physical parameter:

$$x = \frac{r}{r_s}$$

which represents the normalized radial coordinate. Since the relativistic potential is generated as a function of this variable, the model's prediction behavior is directly influenced by changes in (x).

Since the dataset contains a single physical input variable, the feature importance analysis confirms that the model relies entirely on the normalized radial coordinate as the source of predictive information in determining the predicted potential values. This agrees with the underlying mathematical formulation, where the potential is explicitly defined as a function of distance from the gravitational source.

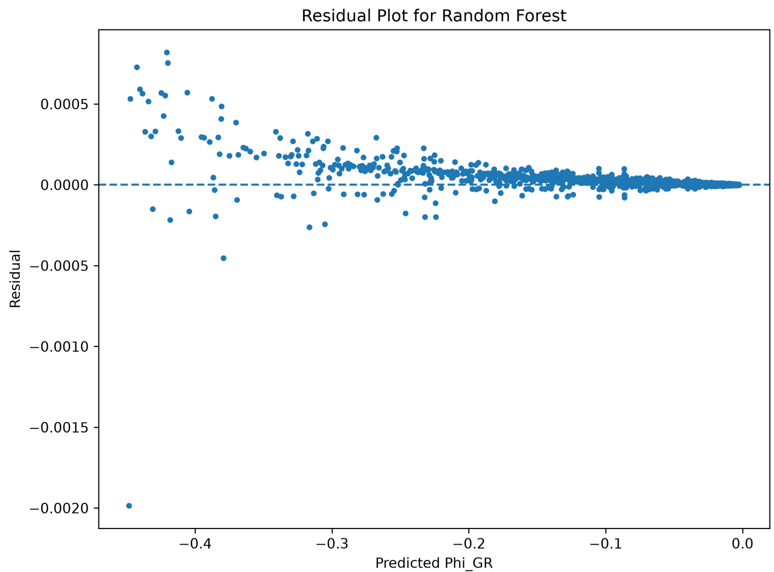

Figure 6: Residual distribution obtained from machine learning predictions, showing the difference between analytical and predicted relativistic potential values.

Although the current model involves a single input feature, the analysis demonstrates the importance of interpretability in scientific machine learning. A high-performing model should not only produce accurate predictions but should also maintain consistency with the physical relationships represented in the data.

The feature importance analysis provides additional confidence that the machine learning models are learning meaningful patterns from the generated dataset rather than producing predictions based on unrelated correlations.

In future extensions involving multiple physical parameters, such interpretability methods can become increasingly important for understanding the contribution of different variables and improving trust in scientific machine learning models.

The contribution of the input variable to the model prediction is illustrated through feature importance analysis.

10 Discussion

10.1 Physics Interpretation

The analytical and computational results provide insight into the relationship between classical and relativistic descriptions of gravitational behavior.

The error analysis demonstrates that the Newtonian approximation captures the qualitative behavior of the relativistic model in the weak-field regime, while

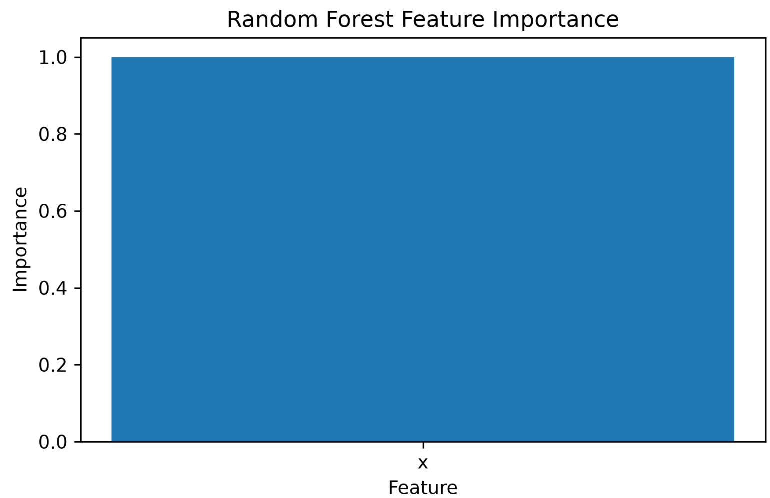

Figure 7: Feature importance obtained from the random forest model. The normalized radial coordinate x is the sole input feature and therefore accounts for the complete predictive contribution.

quantitative differences remain under the chosen normalization. This is consistent with the expected applicability of classical gravitational approximations in regimes where relativistic corrections become sufficiently small.

However, the increasing deviation between the two models in stronger-field regions highlights the limitations of applying classical approximations outside their intended range. The transition between relativistic and Newtonian behavior is not a sudden boundary, but a gradual change that can be quantified through the relative error function.

The computational analysis allows this transition to be studied numerically by identifying regions where the Newtonian approximation achieves specific accuracy levels. This provides a measurable interpretation of the validity range of classical gravity within the considered model.

The results also demonstrate the importance of combining analytical methods with computational approaches. While the mathematical formulation provides the physical foundation of the system, numerical analysis allows detailed investigation of behavior across different regimes.

This combination creates a framework where physical understanding and computational modeling support each other, forming the basis for the machine learning analysis presented in this study.

10.2 Machine Learning Interpretation

The machine learning results demonstrate the ability of data-driven approaches to approximate the behavior of a physically defined nonlinear system.

The comparison between different models shows that the choice of algorithm has a significant impact on prediction performance. Simpler approaches such as linear regression provide a useful baseline but are limited when dealing with the nonlinear structure of the relativistic potential.

The improved performance of neural networks and random forest regression highlights the importance of model flexibility when approximating complex scientific relationships. These models are capable of learning patterns from the generated dataset without explicitly evaluating the analytical expression during prediction.

However, the role of machine learning in this study is complementary rather than a replacement for the physical model. Since the training data itself is generated from the analytical relativistic potential, the machine learning models are evaluated as approximation methods rather than independent methods of discovering the physical relationship.

This distinction is important in scientific machine learning, where models should be evaluated not only by predictive accuracy but also by their connection to the physical system they represent.

The results suggest that machine learning methods can serve as effective computational tools for approximating nonlinear physical models and may be useful in situations where repeated evaluations of complex systems are required. This concept is consistent with the broader development of scientific machine learning, where data-driven models are combined with physical knowledge [9].

10.3 Limitations and Future Extensions

Although the results demonstrate the effectiveness of machine learning methods for approximating the relativistic potential, several limitations should be considered.

The current dataset is generated from a known analytical model with a single primary input parameter. While this provides a controlled environment for evaluating machine learning performance, real physical systems often involve additional variables, boundary conditions, measurement uncertainties, and complex interactions.

Furthermore, the high prediction accuracy achieved by some models is dependent on the quality and range of the generated dataset. A model trained on a specific range of values may not generalize accurately outside that region without additional training data.

Future extensions of this work could involve expanding the model to include additional physical parameters, exploring more complex gravitational systems, and evaluating performance on independently generated numerical simulations or observational datasets.

Another possible direction is the development of physics-informed machine learning approaches, where physical constraints are incorporated directly into the learning process. Such methods could combine the predictive ability of machine learning with the reliability of analytical physical laws. Physics-informed machine learning provides a framework for integrating governing physical principles into machine learning models [9].

These extensions would allow the framework developed in this study to move beyond approximation of a single theoretical relationship toward more complex scientific modeling problems.

The framework developed in this study represents a controlled demonstration of how analytical physics models and machine learning techniques can be combined. While the present system focuses on a single relativistic potential, the methodology can be extended toward more complex scientific problems where exact analytical solutions may be computationally expensive or unavailable.

11 Conclusion

This study presented a combined analytical, numerical, and machine learning framework for investigating a static relativistic gravitational potential and its relationship with the Newtonian approximation.

The theoretical analysis established the mathematical formulation of the relativistic potential and provided a quantitative comparison with the classical model through relative error analysis. The results demonstrated that although the Newtonian approximation reproduces the general inverse-distance behavior of the relativistic model, a systematic quantitative difference remains due to the different asymptotic behavior of the two formulations.

The numerical analysis allowed the deviation between the two models to be investigated across different gravitational regimes. By evaluating the relative error over the selected domain, the study identified the limitations of the Newtonian approximation under the considered formulation and demonstrated how computational methods can provide measurable criteria for approximation validity.

The machine learning component further explored the ability of data-driven models to approximate the relativistic potential from generated physical data. The comparison between linear regression, polynomial regression, neural networks, and random forest regression showed that models with greater capability to represent nonlinear relationships achieved improved prediction performance.

However, the machine learning models were treated as computational approximations rather than replacements for the underlying physical formulation. The analytical model remains the source of the physical relationship, while machine learning provides an alternative method for efficient numerical representation.

Overall, this work demonstrates the potential of integrating theoretical physics, numerical computation, and machine learning for studying nonlinear scientific systems. The developed framework provides a foundation for future extensions involving more complex gravitational models, additional physical parameters, and physics-informed machine learning approaches.

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Finally, I would like to express my profound appreciation to a very special individual whose influence on my personal and academic growth has been immeasurable. At a time when I often underestimated my own potential and struggled with distractions, self-doubt, and uncertainty about my direction, this person consistently encouraged me to aim higher and take my ambitions seriously.

More importantly, this individual demonstrated a level of belief in my abilities that often exceeded my own. Through encouragement, honest conversations, constructive advice, and unwavering support, I gradually began to recognize possibilities within myself that I had previously overlooked. The confidence, discipline, and sense of purpose developed during this period played an important role in enabling me to undertake projects such as the present work.

The research presented in this paper is the result of many months of study, persistence, and effort. However, much of the determination required to begin and continue this journey was strengthened by the support and encouragement received from this individual. Beyond the completion of this project, the influence has extended to broader academic aspirations, personal development, and the pursuit of goals that once seemed beyond reach.

Some contributions to a person's growth cannot be measured through equations, datasets, or performance metrics. Nevertheless, they can leave a lasting impact. For that reason, I offer my sincere gratitude and appreciation for the encouragement, faith, and motivation that helped make this work possible.

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